



Short Communication

Recommender systems in complexity-informed curriculum design: A framework for enhancing personalized learning

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ABSTRACT

This study addresses the persistent limitations of traditional educational recommender systems (ERS), which often fail to model the dynamic, non-linear, and emergent nature of learning. We propose a complexity-informed theoretical framework to reconceptualize the design of these systems, aiming to enhance personalized learning by treating educational technologies as coadaptive ecosystems. To this end, the study addresses four research questions: (1) how the core requirements of personalized learning expose the limitations of static ERS; (2) how principles of complexity theory can redefine their functional architecture; (3) which contextual, behavioral, and pedagogical components most influence their effectiveness; and (4) what algorithmic and design characteristics enable emergent and adaptive behaviors. A structured theoretical review was conducted, involving a systematic search of academic databases from 2008 to 2025 and a two-phase conceptual content analysis. The findings indicate that complexity principles—such as emergence, self-organization, and feedback loops—can fundamentally transform ERS from static filtering tools into dynamic systems that co-evolve with learners. Key characteristics of such systems include adaptive architectures, decentralized processing, and integrated feedback mechanisms. This framework provides a new design logic for creating educational technologies that are more responsive, resilient, and context-aware. Future research must prioritize empirical validation, ensure ethical safeguards such as privacy and transparency, and develop culturally localized models to ensure equitable and effective personalized learning experiences.

1. Introduction

Personalized learning has emerged as a transformative educational paradigm that aligns instructional content, pacing, and learning pathways with learners' cognitive, motivational, and contextual characteristics (Komalawardhana & Panjaburee, 2023; Zhang et al., 2020). In this study, it is conceptualized as an adaptive process in which instructional experiences co-evolve with learners' needs, fostering engagement, agency, and self-regulated learning (Galbally & Christodoulidi, 2023) (see Table 1).

Technological advances and insights from cognitive science have strengthened the need for effective personalization. By integrating learner data, personalized learning seeks to improve instructional alignment while reducing cognitive overload and accommodating diverse learning preferences (Tetzlaff et al., 2021; Zhang et al., 2020). However, existing approaches often overlook cultural and contextual variability, limiting their broader applicability (Morales Murillo et al., 2022).

Recommender systems play a central role in enabling personalized learning at scale by analyzing learner data and generating tailored content and pathways (Sun et al., 2024). Despite their advantages, traditional Educational Recommender Systems (ERS) remain constrained by static user models, data sparsity, and limited capacity to represent nonlinear and evolving learning processes (Tavakoli et al., 2022; Yekollu et al., 2024). Recent comprehensive analyses confirm these persistent gaps across data, algorithmic, and evaluation dimensions (Ma et al., 2024). These limitations often result in rigid and less responsive learning experiences.

Such challenges highlight the need for approaches capable of capturing adaptation, emergence, and continuous change. Complexity theory offers a suitable framework by conceptualizing learning environments as dynamic, coadaptive systems characterized by nonlinearity, feedback loops, and self-organization (Doll, 2008; Guile & Wilde, 2022; Kauffman, 1995).

In this study, curriculum design refers specifically to the dynamic sequencing of learning content. Responding to current limitations, this

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study proposes a complexity-informed theoretical framework for recommender system design in personalized learning environments.

The study addresses the following research questions.

- (1) How do the core requirements of personalized learning expose the limitations of traditional, static recommender systems?
- (2) In what ways can principles from complexity theory—such as adaptability, emergence, and nonlinearity—redefine the functional architecture of educational recommender systems?
- (3) Which contextual, behavioral, and pedagogical components most significantly influence the effectiveness of complexity-informed recommender systems?
- (4) What algorithmic and design characteristics enable recommender systems to exhibit emergent, self-organizing, and context-aware behaviors aligned with complexity theory?

By addressing these questions, the study advances theoretical understanding and informs the development of adaptive and context-aware recommender systems for personalized learning.

2. Methodology

This study adopts a structured theoretical review methodology, suitable for developing conceptual frameworks in interdisciplinary domains (Snyder, 2019). The approach emphasizes theoretical coherence and conceptual synthesis rather than exhaustive coverage.

A systematic search was conducted across five academic databases—Scopus, Web of Science, ERIC, ACM Digital Library, and IEEE Xplore—covering the period from 2008 to July 2025. Two Boolean search strings were applied: (1) ("complexity theory" OR "complex systems") AND ("recommender system" OR "adaptive learning"); and (2) ("personalized learning") AND ("self-organization" OR "emergence" OR "nonlinear interaction").

After duplicate removal, titles and abstracts were screened using predefined criteria. Studies were included if they engaged with complexity theory principles, focused on educational recommender systems, and were published in peer-reviewed venues. Studies lacking theoretical grounding or educational relevance were excluded. This process yielded approximately 120 records, of which 29 were selected based on conceptual relevance and depth. The selection process is summarized in the PRISMA 2020 flow diagram (Fig. 1), generated in accordance with the reporting framework developed by Haddaway et al. (2022).

Data analysis proceeded in two iterative phases. First, inductive thematic coding (Braun & Clarke, 2006) was conducted to identify key constructs across the literature. This process involved repeated cycles of coding and refinement, with each source's conceptual premise as the

unit of analysis.

Second, a theoretical synthesis integrated these themes by examining relationships and tensions between concepts to inform the proposed framework.

To enhance rigor, the analysis was supported by peer debriefing and systematic documentation of analytical decisions.

3. Findings

This section synthesizes conceptual insights derived from the theoretical review to address each of the study's four research questions.

3.1. How do the core requirements of personalized learning expose the limitations of traditional, static recommender systems?

Personalized learning requires instructional approaches that adapt to learners' cognitive, motivational, and contextual conditions while maintaining flexibility, responsiveness, and learner agency (Kirschner, 2017; Rusdi, 2023). These requirements—including dynamic learner modeling, real-time adaptation, ethical transparency, and teacher-system coregulation—reveal structural incompatibilities with conventional educational recommender systems.

Traditional ERS remains limited by.

- static learner profiles that cannot represent fluctuating motivation or conceptual change (Sun et al., 2024);
- linear, historically driven algorithms that fail to capture the nonlinear dynamics of learning trajectories (Zhang et al., 2021);
- data sparsity and cold-start constraints, which inhibit responsiveness to emergent learner behaviors; and
- opaque, black-box models that prevent teacher oversight and hinder pedagogical alignment (Morales Murillo et al., 2022).

Thus, the core requirements of personalized learning—adaptivity, transparency, emergence, and co-agency—highlight structural mismatches in traditional ERS, particularly their inability to respond to fluid, evolving learner behaviors.

3.2. In what ways can principles from complexity theory—such as adaptability, emergence, and nonlinearity—redefine the functional architecture of educational recommender systems?

Complexity theory reframes recommender systems as dynamic, coadaptive ecosystems rather than static filtering tools. A complexity-informed ERS would.

Table 1
Conceptual mapping of complexity theory principles to ERS challenges and complexity-informed design features.

Complexity Principle	ERS Limitation (Challenge)	Complexity-Informed Design Feature (Response)
Nonlinearity	ERS relies on linear, historically driven models incapable of representing evolving motivation or conceptual change; static user profiles and rigid pathways.	Reinforcement learning for dynamic, real-time adaptation to emergent behaviors (Sun et al., 2019; Zhang et al., 2021).
Emergence	Recommendations are predetermined system outputs with no capacity to generate context-dependent, organically arising learning pathways.	Emergent recommendations emerging from iterative learner-system interactions, enabling organic formation of learning trajectories (Kauffman, 1995; Peng et al., 2019).
Self-Organization	Centralized, top-down control over content sequencing limits learner agency and inhibits adaptive learning ecosystems.	Decentralized and federated architectures that enable self-organized learning pathways and distributed decision-making (Lin et al., 2024; Sun et al., 2024).
Feedback Loops	One-directional, opaque systems with no mechanism for learner or teacher input; lack of adaptive refinement.	Bidirectional feedback loops supporting co-adaptation, continuous refinement, and learner/teacher annotation of recommendations (Deldjoo et al., 2020; Deschènes, 2020).
Contextual Embeddedness	One-size-fits-all recommendations that ignore cultural, linguistic, institutional, and infrastructural contexts.	Context-aware modeling, incorporating cultural norms, environmental constraints, and interactional patterns through multi-layered contextual representation frameworks (Sun et al., 2024).

Source: Authors' own illustration.

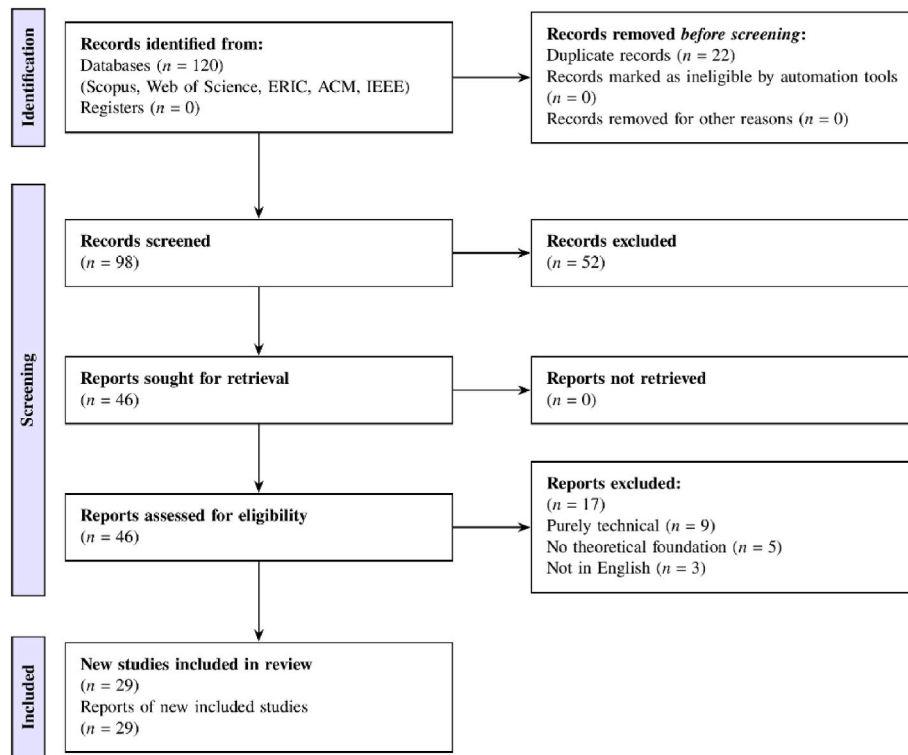


Fig. 1. PRISMA 2020 flow diagram for the systematic search and selection of studies.
Source: Authors' own illustration based on PRISMA 2020 guidelines.

- model nonlinear change by incorporating emergent shifts in motivation, understanding, and behavioral patterns (Peng et al., 2019);
- enable self-organization, allowing recommendations to emerge from iterative learner-system interactions rather than centralized curation (Kauffman, 1995);
- embed continuous feedback loops informed by behavioral traces and social dynamics (Deldjoo et al., 2020; Zhang et al., 2021); and
- scale through decentralized architectures, distributing computation to preserve contextual nuance (Lin et al., 2024).

In this context, AI-driven open learning systems are adaptive digital environments that leverage machine learning algorithms to continuously model learner behavior. Platforms such as Khan Academy exemplify this approach by dynamically adjusting content difficulty and sequencing based on real-time learner interactions. Unlike static filtering, these AI-enabled environments—as discussed by Tavakoli et al. (2022)—integrate behavioral analytics and reinforcement mechanisms to support non-linear personalization.

Recent developments—such as AI-driven open learning systems (Tavakoli et al., 2022) and multi-objective optimization frameworks (Zaizi et al., 2023)—offer promising pathways for operationalizing these principles, enabling ERS design to support dynamic, real-time adaptation to evolving learner behavior and fostering emergent, context-sensitive recommendation patterns.

Overall, complexity theory transforms ERS design from personalization-as-filtering to personalization-as-emergence. This shift reframes recommendations not as fixed algorithmic outputs but as co-evolving responses shaped by learner interactions.

3.3. Which contextual, behavioral, and pedagogical components most significantly influence the effectiveness of complexity-informed recommender systems?

The effectiveness of a complexity-informed ERS depends on three interacting domains: contextual conditions, behavioral dynamics, and

pedagogical foundations.

3.3.1. Contextual conditions

Cultural norms, institutional constraints, linguistic diversity, and infrastructure shape how learners interpret recommendations. Systems that ignore these contextual layers risk producing technically sophisticated but pedagogically misaligned or superficial outcomes (Sun et al., 2024).

3.3.2. Behavioral dynamics

Engagement fluctuates with motivation, self-efficacy, fatigue, and social interaction. Complexity theory interprets these fluctuations not as noise, but as emergent phenomena within learning systems (Harmat & Herbert, 2020). A complexity-informed ERS recognizes disengagement as a transitional phase and adapts support accordingly.

3.3.3. Pedagogical foundations

Teachers remain central as facilitators, co-regulators, and contextual interpreters of learning. Systems that allow educators to annotate, override, or shape recommendations foster coadaptive learning ecosystems aligned with curriculum goals (Guile & Wilde, 2022; Rusdi, 2023).

Thus, system effectiveness emerges not from algorithmic complexity alone, but from integration into the evolving structures of educational practice.

3.4. What algorithmic and design characteristics enable recommender systems to exhibit emergent, self-organizing, and context-aware behaviors aligned with complexity theory?

A theoretically coherent complexity-informed ERS would embody the following characteristics.

- Adaptive learning architectures Online and reinforcement learning models continually update recommendations as learners' behaviors evolve (Sun et al., 2019; Zhang et al., 2021).
- Decentralized data processing Federated and edge-based architectures distribute processing to preserve context and scale efficiently (Lin et al., 2024).
- Hybrid modeling frameworks Combining behavioral traces, semantic content, and social networks captures multidimensional interdependencies (Leskovec et al., 2014).
- Feedback-integrated design Bidirectional loops allow learners and teachers to annotate, accept, or reject recommendations, supporting co-agency (Deschênes, 2020).
- Ensuring robustness through diversity and redundancy Inspired by NK network theory, algorithmic diversity safeguards performance amid uncertainty (Deldjoo et al., 2020). For example, maintaining multiple parallel recommendation models (e.g., both collaborative filtering and knowledge-based approaches) can prevent systemic failures when one model encounters unexpected learner behavior or data shifts.
- Contextual embedding through graph-based structures Dynamic knowledge graphs situate learners within interconnected ecosystems of concepts, peers, and resources (Zhang et al., 2025).

Collectively, these characteristics operationalize complexity theory by enabling recommendations to emerge, self-organize, and adapt within dynamic learning environments.

While these principles are interrelated, each maps onto distinct limitations within ERS, enabling a more coherent translation of complexity theory into system design.

4. Discussion

This study proposes a theoretical framework in which complexity theory functions not merely as a metaphorical lens but as an epistemological foundation for rethinking the design logic of educational recommender systems. Rather than optimizing static forms of personalization, the framework conceptualizes learning technologies as coadaptive ecosystems that evolve through dynamic interactions among learners, content, peers, and pedagogical structures. This approach directly responds to the limitations identified in the Findings section, especially the inability of traditional ERS to accommodate nonlinear learning trajectories, emergent behaviors, and rapidly shifting learner contexts.

4.1. Beyond traditional recommender systems

Conventional educational recommender systems—whether content-based (Aggarwal, 2017; Ricci et al., 2015), collaborative filtering (Gomez-Urbe & Hunt, 2015), or deep learning-based approaches (Zhang et al., 2019)—despite their sophistication, remain grounded in linear assumptions regarding stable preferences, predictable learning pathways, and centralized algorithmic control. Even advanced AI-driven tutoring systems (e.g., Koedinger et al., 2012), although adaptively responsive within circumscribed domains, operate on predefined knowledge structures and treat learning primarily as incremental mastery of fixed competencies. Such systems inherently limit the representation of emergence, self-organization, and contextual fluidity—features that are central to real learning processes.

In contrast, the proposed framework reconceptualizes learning as a nonlinear, self-organizing, and context-dependent process. Recommendations are not predetermined outputs of a fixed model but dynamic, coevolving responses to ongoing learner–system interaction. This shift reframes personalization from algorithmic efficiency toward pedagogical responsiveness, aligning the mechanics of recommender systems with the epistemological assumptions of complexity theory.

4.2. Interpreting the proposed framework

As shown in Fig. 2, a paradigm shift is delineated from traditional educational recommender systems toward complexity-informed architectures. The framework originates in complexity theory principles—nonlinearity, self-organization, and emergence—which directly inspire adaptive system architectures capable of decentralized processing and dynamic modeling.

This progression highlights a fundamental reconceptualization: where traditional systems prioritize stability through fixed user profiles and linear learning paths, complexity-informed ERS embraces dynamic pathways, co-regulation, and real-time feedback. The framework thus moves beyond merely enhancing existing recommendation algorithms toward establishing a new ontological foundation for personalized learning—one where educational trajectories emerge organically through continuous interaction between learners and adaptive systems.

By translating complexity theory into practical design principles, this framework provides a systematic approach for significantly enhancing personalized learning through adaptive, context-aware recommender systems.

4.3. Theoretical contribution and practical implications

This framework contributes to curriculum studies by conceptualizing recommender systems not as technical add-ons but as *curricular actors*—entities that actively shape, rather than merely reflect, the conditions of learning (Guile & Wilde, 2022). It expands complexity-informed curriculum theory by operationalizing core principles such as nonlinearity, emergence, and co-adaptation into concrete system design mechanisms. For educational technology research, the framework offers a design philosophy that prioritizes resilience, transparency, and learner autonomy over predictive optimization alone.

Practically, this reframing implies that teachers are not displaced but augmented as facilitators of emergent learning; data becomes a means for sensing system dynamics rather than profiling learners; and algorithms evolve as transparent, coadaptive partners that support pedagogical judgment. These shifts position personalization as a collaborative process grounded in educational values rather than a purely algorithmic pursuit.

4.4. Limitations and future directions

As a theoretical contribution, this study does not provide empirical validation. Future work must therefore examine the framework in authentic classroom environments. Methodological limitations remain: system performance depends on the diversity and quality of behavioral and contextual data, and the long-term influence of complexity-informed recommendations on deep learning and knowledge retention is not yet known.

Ethically, the framework raises questions that require systematic evaluation, including transparency, explainability, user trust, and vulnerability to data misuse. To advance this line of inquiry, future research should.

- (1) empirically test the framework across varied learning contexts;
- (2) develop evaluation metrics beyond accuracy—such as agency, serendipity, depth of engagement, and resilience;

In this context, *serendipity* refers to an unexpected yet meaningful learning discovery—an outcome conceptually aligned with the classic definition of serendipitous recommendations as "unexpected but useful" (Herlocker et al., 2004; Kotkov et al., 2016). Within complexity-informed learning environments, such discoveries may be understood as emergent phenomena arising from nonlinear learner–system interactions (Kauffman, 1995).

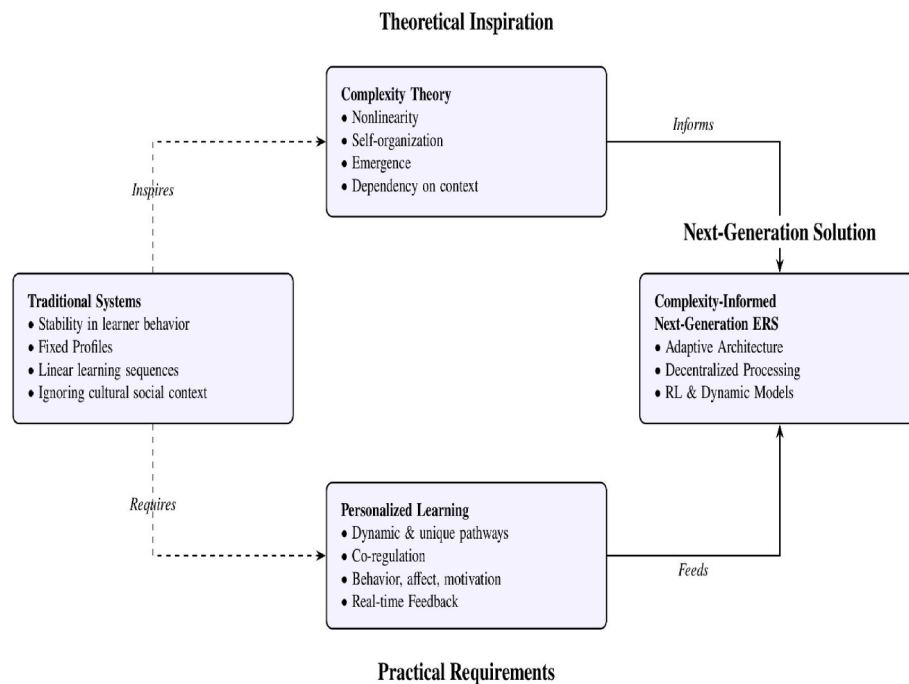


Fig. 2. Conceptual framework for complexity-informed educational recommender systems: Bridging theory and practice.
Source: Authors' own illustration.

- (3) address ethical vulnerabilities, including filter bubbles, privacy risks, and unequal access to adaptive technologies.

Despite these limitations, grounding recommender system design in the epistemology of complexity theory offers a principled pathway toward technologies that *do not merely personalize learning but honor its irreducible, emergent complexity*.

5. Conclusion

This study advances the emerging scholarly discourse at the intersection of complexity theory, curriculum studies, and educational technology by proposing a framework for recommender systems informed by complexity theory, which transcends static and profile-based approaches to personalization. By conceptualizing learning as a dynamic, nonlinear, and coadaptive process, the framework reinterprets recommender systems as participants within the curricular ecosystem—entities that shape learning trajectories through ongoing interaction rather than merely optimizing predefined pathways. This perspective addresses the conceptual limitations identified in the literature and provides a foundation for reimagining the design logic of educational personalization.

By systematically addressing the study's four research questions, we have identified key principles for this evolution. Looking ahead, several interrelated directions can guide future development. First, adaptive algorithms capable of responding in real time to shifts in learner behavior will be crucial for ensuring contextual relevance. Second, enhancing transparency and explainability can foster trust and agency, empowering learners and educators to interrogate and meaningfully engage with system decisions. Third, evaluation practices must move beyond accuracy-based metrics toward indicators that reflect educational priorities, including depth of engagement, serendipity, and long-term knowledge retention. Fourth, culturally localized and pedagogically sensitive models are needed to promote equity and relevance across diverse learning environments. Finally, sustained attention to ethical considerations—particularly privacy, fairness, and autonomy—is essential to ensure that complexity-informed technologies

uphold democratic and humanistic educational values.

In sum, progress in this field depends not solely on more sophisticated algorithms but on interdisciplinary collaboration that integrates pedagogical insight, ethical responsibility, and theoretical rigor. By situating technological innovation within the irreducible complexity of human learning, we can move toward recommender systems that do not merely personalize instruction but respect, support, and co-evolve with learners.

CRediT authorship contribution statement

Fereshteh Hesari: Writing – original draft, Methodology, Investigation, Conceptualization. **Mohsen Ayati:** Writing – review & editing, Supervision, Formal analysis. **Hadi Pourshafei:** Writing – review & editing, Validation, Project administration.

Ethical statement

This research is a theoretical and conceptual study that does not involve human participants, animal testing, or experimental data collection. Therefore, no specific ethical approval was required for this work. All authors confirm that the research was conducted in accordance with academic integrity.

Data availability statement

No datasets were generated or analyzed during the current study.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used generative AI tools solely to assist with language refinement, including improving linguistic clarity, enhancing readability, and refining grammatical structure. These tools were also used to support minor structural streamlining in line with journal formatting requirements.

The authors critically reviewed and edited all outputs generated by

these tools and take full responsibility for the integrity, accuracy, and originality of the content of this publication.

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